STAT 6351: Assignment #5

Notes

This assignment is concerned with alternative approaches to volatility calculation (Lecture 6) and nonlinear models (Lecture 7). Relevant data can be downloaded from Yahoo via the `quantmod` package. For instance:

```
getSymbols('AMZN', from='2007-01-03', to='2017-04-28')
```

Because of possible stock splits, be sure to consider only the adjusted closing prices, whenever the question has you consider closing prices.

Questions

1. (Alcoa stock volatility). Consider the daily log returns r_t of Alcoa stock (tick symbol `AA`), over the period from January 3, 2007 to April 28, 2017. Fit an ARMA-GARCH model to r_t, and plot the resulting (parametric) volatility estimate (estimate of σ_t). Compute also the Moving-Window nonparametric estimate of σ_t (script `mvwindow.R`). Compare both estimates on the same plot.

2. For the daily log returns of Alcoa stock as in Question 1. Compute the Yang and Zhang (2000) nonparametric estimate of the volatility over the entire period (script `yz.R`).

3. (Amazon stock price movement). Consider the daily closing prices of Amazon stock (`AMZN`) and the S&P 500 index, from January 4, 2007 to April 28, 2017. Download the data using `quantmod` from 2007-01-03 to 2017-04-28, and then compute the two series of log returns $\{r_t\}$. Define the *price movement* of Amazon as follows:

$$A_t = \begin{cases} 1, & \text{if } r_t > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Similarly, let M_t denote the *market movement* of the S&P 500 index log returns.

(a) Fit a (linear) logistic regression model for $P(A_t = 1)$ using $\{A_{t-1}, A_{t-2}, A_{t-3}, M_{t-1}, M_{t-2}, M_{t-3}\}$ as inputs. Write down the fitted model. Based on the model, can past price movements of either the stock or the market predict the future direction of Amazon price movement? Why?

(b) Use $\{A_{t-1}, A_{t-2}, M_{t-1}, M_{t-2}\}$ to build a 4-3-1 feed-forward neural network with direct link to model $P(A_t = 1)$.

(c) Divide the sample into training and testing subsamples, with the latter consisting of the last 250 observations. Use 1-step ahead predictions $\hat{A}_t$ to compare the models in (a) and (b). Since predictions must be binary, let $\hat{A}_t = 1$ if $P(A_t = 1) > 0.5$, and zero otherwise. (Note: the script `backnnet.R` will perform backtesting for neural nets, just as `backtest.R` does for ARMA models.)